
DADS 6002 / CI 7301

Big Data Analytics

Hive Lab

Hive Lab

- Start Hive
hive
hive >
- Exit from Hive
hive > exit;
- Create a database
hive> create database my_db;
hive> show databases;
hive> use my_db;

Hive Lab

- Create and alter a table

```
hive > create table test ( id int, name string ) row format
delimited fields terminated by ',' stored as textfile;
hive > show tables;
hive > describe test;
hive > alter table test add columns ( address string );
hive > describe test;
hive > drop table test;
```

Hive Lab

- Download a zip file ml-100k.zip from MS Teams into the shared folder and copy to the working directory (or download it from the web as follow.

```
# wget http://files.grouplens.org/datasets/movielens/ml-100k.zip )
```

```
# unzip ml-100k.zip
```

```
# more ml-100k/u.user
```

Hive Lab

```
# cd ml-100k
# hadoop fs -mkdir /user/cloudera/movielens
# hadoop fs -put u.user /user/cloudera/movielens
# hadoop fs -ls /user/cloudera/movielens
# hive
hive > create table users ( userid int, age int,
    gender string, occupation string, zipcode string )
    row format delimited fields terminated by '|'
    stored as textfile;
```

Hive Lab

```
hive > load data inpath
    '/user/cloudera/movielens/u.user' overwrite into
    table users;

hive > select * from users;
hive > select  zipcode, count(1) as count, avg(age)
    as age from users
    group by zipcode order by count desc;
hive > quit;
```

Hive Lab

- Download a file wlog from MS Teams into the shared folder and copy it to the working directory
- Execute the following commands.

```
# hadoop fs -mkdir /user/cloudera/weblog
```

```
# hadoop fs -put wlog /user/cloudera/weblog
```

```
# hadoop fs -ls /user/cloudera/weblog
```

Hive Lab

```
hive> create table weblogtest ( host string, object string,  
    time string )
```

```
ROW FORMAT SERDE
```

```
'org.apache.hadoop.hive.contrib.serde2.RegexSerDe'
```

```
WITH SERDEPROPERTIES
```

```
("input.regex" = "([^ ]+) \"([^ ]+)\" ([0-9]+)" ) stored as  
textfile;
```

Hive Lab

```
hive > load data inpath '/user/cloudera/weblog/wlog'
      overwrite into table weblogtest;
hive > select count(1) from weblogtest;
hive > select object, count(1) as count from weblogtest
      group by object order by count desc;
```

Hive Lab

```
hive> create external table weblog (
      host string,
      object string,
      time string )
ROW FORMAT SERDE
  'org.apache.hadoop.hive.contrib.serde2.RegexSerDe' WITH
SERDEPROPERTIES
  ("input.regex" = "([^ ]+) \"([^ ]+)\" ([0-9]+)" ) stored as textfile location
  "/user/cloudera/weblog";
hive > select object, count(1) as count from weblog group by object
      order by count desc;
```